Clinical Trials in Global Health 5

Reporting summary results in clinical trial registries: updated guidance from WHO

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The importance of publicly registering clinical trials and reporting their results in registries is widely recognised. While substantial progress has been made with registering trials before enrolment, the availability of results in registries remains uncommon despite expanding legislative and funder requirements—leading to an incomplete evidence base and avoidable waste of resources, particularly for unpublished trials. This paper discusses the rationale for reporting summary results in trial registries, reviews the current landscape of registry policies, and presents new WHO guidance for reporting results in registries. The 2025 WHO guidance was developed after consultation with relevant parties, including researchers, patients, sponsors, funders, regulators, journal editors, registry administrators, and the public. The guidance defines eight minimum items that are essential for understanding and interpreting the summary results for all trials. Implementation of the WHO guidance by trial registries, broad adherence by investigators and sponsors, and endorsement by funders, regulators, legislators, research ethics committees, patient organisations, and journals can help enhance the contribution of trials to scientific knowledge, patient care, and health policy.

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Introduction

Public registration of clinical trials, and the timely and complete reporting of their results are internationally endorsed mechanisms to increase the value and effect of trials.[1] The public availability of information about the methods and results of all trials helps sponsors and investigators to fulfil their ethical responsibilities to study participants; informs researchers, patients, governments, and policy makers about the global trials landscape; and promotes a complete evidence base that is not skewed by the selective reporting of favourable results. There has been substantial progress over the past two decades in registering most trials. Conversely, the results of most trials remain unreported in registries. In this paper, we discuss the rationale for reporting summary results in trial registries, review the current landscape of registry policies, and present new WHO guidance for reporting results in registries.

Importance of trial registration and results reporting

Systematic reviews of clinical trials help to inform health-care providers, patients, researchers, and policy makers about the benefits and harms of interventions, and provide directions for new research and development. The validity of these evidence syntheses depends on the availability of results that are representative of the global body of evidence. Transparent and full reporting of the methods and results of all trials has been widely endorsed as an ethical and scientific imperative to promote a complete evidence base and reduce avoidable research waste,[1–3] particularly because statistically significant or favourable results are often selectively reported in journals over statistically non-significant results. Research waste refers to the poor return on investment of resources—including funding and spent by research teams and trial participants—trials do not contribute to scientific knowledge, good, and patient care. Ensuring that trials public knowledge also helps sponsors and to fulfil their ethical responsibilities to study who voluntarily enrol in trials and accept in exchange for scientific progress.[2,4]

Key recommendations and proposed metrics for reporting summary results in registries

- For every clinical trial, the sponsor should report summary items recommended in the 2025 member registry of the WHO 12 months of trial completion
  - Metric: the available in a public registry trial completion date
- Governments and registry registries are adequately 2025 WHO guidance for including the use of structured
  - Metric: the results-related data fields guidance
- Funders, regulators, legislators, committees, and journals should policies that require the reporting accordance with the 2025 WHO
  - Metric: the legislators, research ethics with applicable policies

Trial registration

Prospective registration of clinical trials in a public registry is a central mechanism to increase transparency about the cumulative evidence base for health-care interventions and policies. Registration is considered prospective when it is completed before participant enrolment begins, a central criterion for a trial to be considered adequately registered.[2,5] When universally implemented, the prospective registration of basic trial information serves to document the existence of every trial along with its planned outcome measures.

Public availability of registered information has several benefits. Researchers and policy makers can use registries to conduct meta-research and evaluate trends and gaps in the global landscape of trials for a given intervention, health condition, or geographic region.[6] For example, an analysis of trial registry data revealed that over 2000 trials of drug therapies for COVID-19 were launched from March to October, 2020, but only 5% were randomised and adequately powered to yield informative results.[7] Patients can search registries to identify relevant trials available for their participation.[8] For systematic reviews, trial registries can be an important source of information about ongoing or unpublished trials.[9]

By providing a record of pre-specified outcomes, registries are also a key source of information for assessing the risk of bias for trials.[10] The ability to compare the pre-specified outcomes, as defined in the date-stamped registry record, with the reported results of completed trials provides a means to deter and identify the selective reporting or omission of outcomes.[11] Trials that are prospectively registered are more likely to be published without outcome discrepancies,[12] demonstrating the value of prospective trial registration in promoting accurate and transparent reporting. Beyond documenting trial outcomes, registries have been

non-publication, publication bias, and outcome reporting

shown in some cases to provide more methodological information than journal publications.[13]

Results reporting in registries

The reporting of favourable and unfavourable results from all trials helps to ensure that an unbiased and complete body of evidence is available to inform patient care, research, and health policy. However, one fifth to one half of clinical trials remain unpublished within 2 years or 5 years of trial completion, or 10 years of ethics approval, depending on the sample.[1,14–16] Even when published, about a third of trials have discrepancies in the primary outcomes defined in publications versus registry records or protocols, with statistically significant or more favourable outcomes being reported over non-significant ones.[11,17–19] The selective publication of trials with statistically significant results distorts the available literature, leading to suboptimal care and, in some cases, substantial harm to patients.[1]

The reporting of results in registries can help to reduce

bias. For unpublished trials, registries can be the only public source of information about results. Meta-research studies reviewing trials conducted in Switzerland, Germany, the UK, Canada, and the USA found that 29–40% of unpublished trials had results available in trial registries.[14–16,20] Other studies have shown that the results reported in trial registries can be more complete compared with journal publications, particularly for harms data.[21] These findings highlight the importance of searching registries as part of systematic reviews.[9,22,23]

Progress of trial registration

Over the past two decades, an increasing number of governments, regulators, journal editors, and funders have implemented policies to promote trial registration within their constituencies. A major catalyst to promote trial registration was the International Committee of Medical Journal Editors' decision to require trial registration as a condition of publication in their member journals starting in 2005.[24] Alongside this major policy implementation, a 2005 World Health Assembly resolution called for the creation of a platform to link trial registries and ensure unambiguous identification of trials, leading to the launch of the WHO International Clinical Trials Registry Platform (ICTRP) in 2006.[25,26] The 2008 revision to the Declaration of Helsinki solidified the ethical imperative of registration by adding requirements for all trials to be publicly registered before recruiting participants.[27]

Since 2006, the ICTRP has led the development of international norms and standards for trial registration and registries. The WHO Trial Registration Data Set defines the minimum information about a trial that should be publicly registered before enrolling participants.[2,28] In addition to administrative details, the Trial Registration Data Set includes basic information about the trial design, including the pre-specified outcomes and target sample size. This Trial Registration Data Set allows the existence and reporting of trials to be tracked from their inception and has been widely adopted by trial registries, medical journals, and legislators.

Another key role of ICTRP is the coordination of a global network of trial registries. The WHO Registry Network currently includes 20 national and international registries that regularly upload trial data to the ICTRP search portal database,[29] which enables access to data aggregated across registries. 17 of these data providers are primary registries that are accountable to meet ICTRP-defined standards related to content, quality, accessibility, technical capacity, and administration. Although registries can maintain protocol records in their national language, data are exported to the ICTRP database in English in a standardised, downloadable format. This standardisation facilitates visibility and analysis of the landscape of trials conducted around the world.

avoidable research waste and patient harms arising from

The growing implementation of ICTRP policies and widespread adherence mechanisms by journals,[30,31] funders,[32] and regulators[33] has translated into concrete

the 2024 WHO guidance for best practices for clinical trials highlighted the need for timely, complete, and accurate reporting of results in a publicly available format.[41,46]

advances in trial registration. As of May, 2024, over one million study records are accessible in the ICTRP database, as contributed by member registries of the WHO Registry Network. Over 14 000 trials were evaluating interventions related to COVID-19. The number of studies registered annually has almost tripled from 26 000 in 2006, to 72 000 in 2023.

Although substantial progress has been made, there

Despite growing support from funders and legislators,[47] adherence remains low in terms of timely reporting of summary results in registries. Reviews of registry records found that fewer than half of regulated trials had results uploaded to ClinicalTrials.gov or the European Union Clinical Trials Register, despite being mandated by legislation.[48–51] Reporting of results in registries would remain opportunities for improvement in trial registration practices and registry infrastructure. Existing adherence mechanisms apply to a subset of trials and are not consistently enforced, leading to some trials not being be expected to be registered—particularly those evaluating non-regulated interventions (ie, not drugs or devices).[34] Among trials receiving ethics approval in Switzerland, the UK, Germany, and Canada in 2012 or between 2016 and 2020, 16–31% were not prospectively

regulated interventions (eg, surgery) that fall outside of the scope of current legislation.[34] Implementation of the 2017 WHO Joint Statement has also been variable among funders.[52,53] Overall, this means that although searches of trial registries often help to identify additional relevant but unpublished trials for systematic reviews, most of these trials cannot inform patient care or be included in meta-analyses because their results have not registered.[14,35] In reviews of registry records, up to half of been reported.[9,54] trials were submitted late to registries (ie, after participant enrolment),[14,36–40] contrary both to internationally accepted ethical requirements[41] and to the scientifically important intent of publicly documenting the pre-specified trial outcomes and target sample size before a trial begins. Furthermore, the outcomes defined in a registry record might not match those specified in the protocol or other registry records for the same trial.[12,42,43] These gaps are compounded by the variable adherence of registries to ICTRP-defined minimum standards for data accessibility, content, audit trails, web security, and user guidance.[44]

Progress of reporting results in registries

The value of trial registries is derived from providing not only a date-stamped record of basic protocol information before participant enrolment, but also a mechanism to provide public access to the summary results. If trial results are not made publicly available, then the investment of resources and efforts to complete the trial is wasted. The 2015 WHO Statement on Public Disclosure of Clinical Trials Results defined a 12 month timeframe for posting the final enrolment number, completion date, and results for key outcomes in registries.[45] In 2017, 21 national and international health research funders signed a WHO Joint Statement on the Public Disclosure

Mapping the landscape of registry items relevant to results

From 2022 to 2023, we reviewed the websites of ClinicalTrials.gov and 16 WHO primary registries for publicly available information on the data elements they collected that were relevant to summary results. We included data elements that were captured as either optional or mandatory fields. We also contacted each registry by email (n=16 responses) to confirm the information found and to solicit any missing information. One current WHO primary registry—the International Traditional Medicine Clinical Trial Registry—was not included in the Review, as it had not yet been established. There was substantial variation in the type and format of results-related information collected by the 17 registries after trial completion (table 1). This information includes elements of trial methods that are relevant to interpreting and appraising the results. Overall, the registries recorded a total of 42 unique elements related to summary results. The most commonly recorded elements across registries were related to the description of trial participants, key trial dates (ie, start of enrolment, trial completion, and first journal publication), URL link to publications, and plans for sharing individual participant-level data. Two specific elements—participant flow and baseline

of Clinical Trial Results summary results to be reported in trial registries within 12 months of study completion; two additional national funders became signatories in 2020.[32] To further support results reporting efforts, WHO expanded the Trial Registration Data Set in 2017 to include an item on summary results, outlining general items that should be reported in registries. In addition, legislation in several countries requires summary results to be posted in registries for regulated trials. Most recently, the 2024 revision to the Declaration of Helsinki and

characteristics—were recorded by all 17 registries, while 24 elements were captured by fewer than half of the registries. The least commonly recorded elements were related to study limitations, conflicts of interest, statistical methods, and adverse event grading and assessment. Registries collected information in various formats, including structured data fields (n=4 registries) and free text or document uploads (n=13).

The variability across registires largely reflects the absence of harmonised guidance for the specific

	ANZCTR	ChiCTR	ClinicalTrials.gov	CRiS	CTRI*	DRKS	EUCTR	IRCT	ISRCTN	jRCT	LBCTR	PACTR	ReBEC	REPEC	RPCEC	SLCTR	TCTR
Completion status																	
Whether	X	X	X		X	X	X	X		X			X				
Actual	X	X	X	X	X	X		X		X	X		X	X	X		
Actual	X	X	X	X	X	X	X	X		X	X			X	X	X	
Participant flow																	
Table or	X	X	X	X	X	X	X	X	X		X	X	X	X	X	X	X
CONSO	X	X				X	X	X					X		X	X	
CONSO			X	X	X				X			X		X			
Baseline characteristics																	
Any	X	X	X	X	X	X	X	X	X		X	X	X	X	X	X	X
Age†		X	X	X	X		X		X			X	X		X	X	X
Sex or		X	X	X	X		X		X			X	X		X	X	X
Race or		X	X		X											X	
Final outcome definitions																	
Specificatio		X	X	X		X	X	X		X			X	X	X		X	X
Description		X	X	X		X	X	X		X							X	X
Method of		X	X	X			X	X										
Timepoint†		X	X	X		X	X	X		X							X	X
Outcome results																	
Summary instructi									X		X			X		X	X	X
Overall	X	X	X	X	X	X	X	X							X			
Type of		X	X	X	X		X	X							X			
Summar	X	X	X	X	X	X	X					X	X		X			
Effect	X	X	X	X	X	X	X					X	X		X			
Measure	X	X	X	X	X	X	X	X				X	X		X			
Names		X	X		X		X	X				X	X		X			
Subgrou	X	X	X		X	X	X											
Statistical methods																	
Framework (eg,			X	X			X		X							X		
Type of hypothesis			X	X			X		X	X				X		X	X	
Reasons for			X	X			X									X		
Handling of			X	X			X											
Harms or adverse events																		
Summary instructi	X					X					X		X	X			X	X
Any		X	X	X	X		X		X	X					X			
Serious		X	X	X	X		X		X	X					X			
Deaths†		X	X	X	X		X		X						X			
Timefra		X	X	X	X		X								X			
Approach systemat		X	X		X		X											
Severity	X	X	X		X	X												
Number of	X	X	X		X	X	X								X	X		
Trial summary																	
Scientific		X						X		X	X			X			X	
Public	X	X		X	X	X					X	X				X		X
Limitations																	
Study limitations						X				X						X		
Conflicts of interest†						X				X				X				
Reasons for delayed disclosure of						X				X								

(Table 1 continues on next page)

	ANZCTR	ChiCTR	ClinicalTrials.gov	CRiS	CTRI*	DRKS	EUCTR	IRCT	ISRCTN	jRCT	LBCTR	PACTR	ReBEC	REPEC	RPCEC	SLCTR	TCTR
(Continued from previous page)																	
Study materials																	
Study	X	X	X	X		X	X		X	X	X	X		X	X		
Protocol	X	X	X			X				X				X	X		
Statistical	X	X	X			X			X								
Informed	X	X	X			X			X								
Publications																	
URL or	X	X	X	X	X	X	X	X	X	X	X	X	X		X	X	
Date of	X	X	X		X	X		X	X	X	X	X	X	X	X	X	
Participant-level data																	
Willingnes	X	X	X		X	X	X		X	X	X	X		X	X	X	
Description	X	X	X		X	X	X		X	X	X	X		X	X	X	

ANZCTR=Australian New Zealand Clinical Trials Registry. ChiCTR=Chinese Clinical Trial Registry. CRiS=Clinical Research Information Service (Republic of Korea). CTRI=Clinical Trials Registry India. DRKS=German Clinical Trials Register. EUCTR=EU Clinical Trials Register. IRCT=Iranian Registry of Clinical Trials. ISRCTN=International Standard Randomised Controlled Trial Number. jRCT=Japan Registry of Clinical Trials. LBCTR=Lebanese Clinical Trials Registry. PACTR=Pan African Clinical Trial Registry. ReBEC=Brazilian Clinical Trials Registry. REPEC=Peruvian Clinical Trials Registry. RPCEC=Cuban Public Registry of Clinical Trials.
SLCTR=Sri Lanka Clinical Trials Registry. TCTR=Thai Clinical Trials Registry. *Information collected by registry, but not yet made public. †Included in new WHO guidance on reporting results.

elements to be collected after trial completion. While the widely endorsed CONSORT Statement defines a minimum set of items to report in publications of completed trials,[55] specific recommendations are lacking for reporting results in trial registries. This gap in the global trial ecosystem means that diverse data elements are currently captured in various formats that cannot be easily exchanged or aggregated electronically across trials and registries. Accordingly, the WHO ICTRP search portal database currently does not include data relevant to summary results, aside from a flag indicating whether

comments from 80 interested parties in 23 countries. After considering the diverse feedback, the steering group iteratively revised the list of potential items in consultation with members of the ICTRP Registry Network.

The 2025 WHO guidance defines eight minimum items relevant to summary results that should be reported in registries (table 2). The items relate to the full protocol, trial completion status, key dates, description of participants, outcome results for benefits and harms, and conflicts of interest. In line with the 2015 WHO Statement and 2017 Joint Statement, these items should

a trial has results available. be reported in a registry within 12 months of trial

2025 WHO guidance for reporting summary results in registries

In consultation with relevant parties from 2022 to 2024, the WHO ICTRP led the development of harmonised guidance for reporting results in registries. The objectives were to facilitate implementation of the 2017 WHO Joint Statement and help registries capture a consistent set of items relevant to results reporting. To inform the new standards, the range of data items currently recorded by trial registries was mapped (ie, in table 1), along with applicable items recommended by the CONSORT Statement.[55] Based on these reviews, a preliminary list of potential results-related registry items was drafted. An expert consultation group reviewed the draft list by email and discussed the items during an online meeting in July, 2022. The expert consultation group consisted of 41 individuals and organisations from 13 countries who represented the perspectives of researchers, patients, methodologists, journals, funders, regulators, contract research organisations, and industry sponsors. The meeting was followed by an open public consultation conducted online in November, 2022 that received written

completion, defined as the date of the last follow-up assessment for the primary outcome.

Scope

The 2025 WHO guidance on reporting summary results in registries applies to all registered clinical trials, including those that were stopped earlier than expected. The guidance focuses on the minimum set of information required to understand and interpret the summary results in registries for any trial design.

The recommended registry items are based on a subset of items from the widely endorsed CONSORT guideline for reporting completed trials in journal publications.[55] The information reported in registries complements the information reported in journals.[14–16,21] Together, these data sources within the trial reporting ecosystem facilitate an appropriate interpretation and appraisal of trial methods and results. Journal publications adhering to the CONSORT guidance provide peer-reviewed, narrative insights that are not available in registries. Summary results can typically be made publicly available sooner in registries than in journal publications, but registry data are not peer-reviewed. For the substantial proportion of

	Item name	Description	Components
1	Trial protocol	Most recent study protocol and full statistical including version number, date, and history	Document upload or digital object identifier
2	Completion status	When and why the trial ended or was stopped	Completion date and completion status
3	Dates of reporting results	Dates when results were reported in a journal	Date and digital object identifier of first to the registry; and date of posting of results
4	Participant flow	Progress of participants from study enrolment analysis (ie, based on CONSORT flow diagram)	For each group, numbers of participants who intervention as allocated (with reasons), were analysis of primary outcome (with reasons)
5	Participant characteristics	Characteristics of participants at baseline	For each group, descriptive summary sociodemographic and study-specific
6	Outcome results	(a) Definition of each primary and secondary outcome	Specification (eg, primary outcome); method of aggregation (eg, mean); and
6	Outcome results	(b) Who is included in the analysis, and in outcome	Definition of analysis population (eg, all group)
6	Outcome results	(c) Summary by group for each outcome and analysis population	Name of trial group; number of participants with the outcome event; and for continuous measure of variability (eg, standard deviation
6	Outcome results	(d) Comparison between groups for each population	Names of groups being compared; effect difference, and difference in means); effect and p value (optional)
7	Harms or adverse events	Unfavourable changes in health (eg, new or symptom, abnormal laboratory finding) in regardless of causal relation to the study	Definition of harms analysis population (eg, assigned group); and numbers and adverse events (overall and each type), and
8	Conflicts of interest	Financial and non-financial relationships that of interest	Any conflicts of interest for trial steering publication)

Table 2: **Minimum items recommended by WHO for reporting summary results in trial registries**

trials that remain unpublished, a registry might represent the only public source of information about their methods and summary results.

Explanation of recommended items

Item 1: trial protocol

The trial protocol and full statistical analysis plan, including all amendments, provide the most comprehensive descriptions of the study design and methods. Access to these documents is essential to provide appropriate context and allow an understanding of how the results were generated. For example, the interpretation of outcome comparisons between groups (ie, item 6d) will differ depending on whether the conceptual framework of the trial was designed to evaluate superiority, non-inferiority, or equivalence.

The protocol and full statistical analysis plan should be available in the registry alongside the trial results.[56] It is important to maintain version control for these key documents by including the version number, date, and list of amendments made in each version. Guidance exists to promote complete and clear content by defining the essential items to address in protocols and full statistical analysis plans.[57,58]

Item 2: completion status

The expected end of a trial typically occurs when the overall target number of enrolled participants has completed follow-up, the target number of outcome events is observed, or a set time period or date is reached.

Some trials might stop earlier than expected due to feasibility issues, harms, futility, or benefit.[59,60] Reporting the completion status and the reason for stopping early provides important context for interpreting the results and understanding why there are less data available than originally planned. For example, there is potential for overestimation of treatment effects when a trial is stopped early for benefit.[61] Even if there are insufficient data to conduct the protocol-specified analysis, reporting of outcome data on benefits and harms enables their inclusion in meta-analyses and can help inform future research. The availability of results in registries is particularly important when trials are stopped early, as they are less likely to be published in journals than trials that are completed as expected.[62]

Item 3: dates of reporting results

A growing number of research funders, organisations, and governments have introduced policies mandating the public posting of results within specific timeframes for trials under their remit. The 2015 WHO Statement and 2017 Joint Statement on results disclosure called for the reporting of results in registries within 12 months and in journals within 24 months after trial completion—defined as the date of the last follow-up assessment for the primary outcome. Monitoring adherence to these policies requires a public record of key dissemination milestones, including the dates of submission to a registry, public posting of the information after review by the registry, and publication of the main results in a journal.

Item 4: participant flow

The extent of participant adherence to the planned intervention and follow-up protocols has implications for a trial's interpretation and risk of bias. Low adherence to interventions might reduce the ability to identify any benefits or harms. Substantial losses to follow-up and exclusions from analyses can introduce bias—particularly if the number lost or excluded differs between trial groups—because non-adherers might have a different prognosis than adherers. Exclusion of participants also reduces the statistical power of analyses. A review of ClinicalTrials.gov records in 2010 found that a quarter of trials excluded at least 10% of enrolled participants from

(ie, intention-to-treat analysis) is least susceptible to

Some analyses exclude specific participants, such as those with missing data or those who do not meet a specific level of adherence to the interventions. The implications of the choice of analysis population can only be understood if registries record an explicit definition of who is included in the main analysis for each outcome.

Item 6c: outcome results—summary of each group

The outcome data for the main analysis population can be summarised by trial group in various ways, depending on the type of data. For all outcomes, the number of participants in the analysis should be reported by group.

the analysis.[63] **For binary outcomes, it is sufifcient to also report the**

The implications of exclusions and losses to follow-up can be better understood with clear reporting of the progress of participants from enrolment to completion in each trial group—including the number who enrolled, did not receive the planned interventions, and were lost to follow-up, along with reasons for exclusion from the main analysis population (ie, item 6b) for the primary outcome. This information aligns with recommendations from the CONSORT 2025 Statement.[55]

number of participants in each trial group who have the outcome event. For continuous outcomes, the average value (eg, mean or median) and a measure of variability (eg, standard deviation or interquartile range) can be reported to summarise the data for each group.

Item 6d: outcome results—comparisons between groups

Statistical comparisons between trial groups estimate the effect of the intervention on each outcome. The effect measure (eg, relative risk or difference in means), effect size, and its confidence

Item 5: participant characteristics

A summary of the baseline characteristics of trial participants provides insight into the applicability of the trial results to specific patients. Descriptive summary statistics (eg, proportion or mean with measure of variability) should be reported by trial group for sociodemographic variables, including age and sex,[64] and other characteristics relevant to the health condition, interventions, and outcomes being studied.

precision) should be reported for each outcome. It is also necessary to specify which groups are being compared, particularly for multi-group trials.

Item 7: harms or adverse events

Harms or adverse events refer to any unfavourable changes in health arising during the trial. These harms might be causally related to the trial interventions, and can be collected systematically (eg, questionnaire) or non-systematically (eg,

Item 6a: outcome results—definitions

As part of the WHO Trial Registration Data Set, the pre-specified primary and secondary outcomes are defined in registries before participant enrolment. Results should be reported for each of these registered outcomes. Key components of an outcome include the specific

events can be characterised by their frequency, severity grade, seriousness, and causal relation to the intervention.[65,66] Although single trials are typically not powered to detect differences in adverse events, full reporting of harms data allows for meta-analysis across

measurement variable (eg, Psoriasis Areas and Severity trials.

Index-90), analysis metric (eg, final value or change from baseline), method of aggregating individuals' data for each trial group (eg, mean or proportion), and timepoint of interest.[63] Any changes to outcome definitions or the choice of primary or secondary outcomes should be transparently documented in registries so that it is clear which outcomes were not pre-specified and why the change was made. Selective reporting or non-reporting of outcomes can introduce bias if performed based on

Registries provide an opportunity to report a complete record of adverse events, as the public reporting of harms data is often incomplete in journal publications and typically absent for unpublished trials.[67] As with all trial outcomes (ie, item 6b), it is important to define which participants are included in the analysis of harms. Key data to report for each trial group include the number and proportion of participants experiencing death, serious adverse events, any adverse event, withdrawals

the results.[11,17,18] due to harms, and each specific type of adverse event

Item 6b: outcome results—analysis population

(eg, diarrhoea) observed during a trial. These data should also be reported by severity grade.

The results can vary depending on the types and numbers of participants who were excluded in each analysis (item 4).[63] An analysis population that includes all enrolled participants in their original allocated group

Item 8: conflicts of interest

Industry sponsorship and financial ties of principal investigators have been shown to be associated with the

publication of more favourable efficacy results and conclusions compared with a lack of industry relationships.[68,69] Aligned with standard requirements for publishing in journals, the declaration of relevant financial and non-financial relationships facilitates an understanding of conflicts of interests and their potential effect on the conduct and reporting of a trial. At a minimum, any conflicts of interest for members of the trial steering committee should be reported, given their key role in overseeing the trial.

Limitations

The new 2025 WHO guidance focuses on the minimum items to include in registries rather than an exhaustive list, recognising the need to balance the quality and completeness of registry fields with the burden of

the 2025 WHO guidance. This includes resources for improving infrastructure to meet existing ICTRP-defined standards, for incorporating the new results-related fields in their database, and for conducting quality control of submitted data. Investment in registry infrastructure and personnel is also essential to promote the flow of high-quality structured data between registries and the ICTRP database. Implementation of the WHO guidance can be monitored based on the proportion of registries that allow submission and public posting of all results-related fields.

It is recommended that registries offer structured data formatting for applicable items. Structured data fields with defined response options allow for more **comprehensive and efficient search queires and analyses** compared with unstructured, free-text fields or document uploads. Machine readability also facilitates quality

registering results. Several important elements were checks for errors.

ultimately excluded during the iterative consultation and development process. Specific items related to the data collection methods (eg, assessment of harms), conceptual framework (eg, superiority), and statistical analysis methods (eg, type of hypothesis test and handling of missing data) were excluded because they should already be detailed in the trial protocol and statistical analysis plan that will be submitted to the registry as part of item 1. Other important items (eg, public summary, scientific abstract, or limitations and caveats) were not retained in the guidance due to the potential for inappropriate promotional language in a narrative text field that would not be peer reviewed.

The WHO guidance is restricted to the reporting of summary results and does not address the sharing of individual participant-level data. WHO supports efforts to promote data sharing as an important component of Open Science efforts to encourage research transparency, accessibility, and reproducibility. The WHO Trial Registration Data Set currently includes an item about whether and how participant level data will be shared. These data sharing plans submitted with the initial registration should be updated as needed upon trial

Third, relevant parties with adherence mechanisms—including funders, sponsors, regulators, legislators, **research BM quantum computers leverage furniture committees, and journals—should** implement policies requiring adherence to the 2025 WHO guidance on reporting summary results in registries. The substantial increase in adherence to trial registration over the past two decades was only possible due to the complementary actions of these interested parties. A similar effect could be achieved for results disclosure if the organisations and governments that fund, sponsor, oversee, and publish trials introduce and enforce policies aligned with the WHO recommendations. The prevalence of such policies would serve as an informative metric for monitoring progress.

Over the coming years, ICTRP will work closely with registries to operationalise and implement the 2025 WHO guidance for reporting summary results. The recommended items will be incorporated into the WHO Trial Registration Data Set to create the WHO Trial Registration and Results Data Set—recognising that some items (eg, outcome definitions) will have been submitted before enrolment as part of the initial registration record and will need to be confirmed or

completion. updated by registrants upon trial completion. Consistency

Recommendations for implementation

To achieve widespread reporting of summary results in registries for all clinical trials, we propose three key recommendations along with metrics to monitor adherence. First, investigators and sponsors should ensure that summary results are posted in a trial registry within 12 months of study completion. Adherence to this recommendation can be evaluated in terms of content and timeliness based on (1) the proportion of registered trials that make available in the registry the eight results-related items defined in the WHO guidance; and (2) the proportion that do so within 12 months of the date of the last data collection for the primary outcome.

Second, governments and registry funders should provide sufficient resources for registries to implement

in the items common to the WHO guidance and the CONSORT 2025 reporting guideline will make it easier for trial investigators and sponsors to submit the same information to the registry and journal.

To increase the value derived from trials, it is crucial that relevant parties adopt policies to support and endorse the 2025 WHO guidance for reporting summary results in registries for all trials. Building on the substantial progress with trial registration, ICTRP will work with registries, researchers, funders, patient organisations, regulators, governments, journals, and other relevant parties to promote broad adherence to the new WHO guidance. Aligned with ongoing WHO activities to implement World Health Assembly resolution 75.8 (2022) to strengthen global clinical trials capacities,[46,70,71] these efforts will help ensure that the

efforts of trial participants, sponsors, and
produce publicly available data that can
improved health care.

Contributors

A-WC, GK, and VM were responsible for the concept. A-WC, GK, LRdS, LH, ALR, CMT, and VM were steering group guideline development. JP, GK, and A-WC conducted the survey of results-related items in registries. All authors comments from the consultations with external parties. A-WC the initial draft of the manuscript. All authors then critically manuscript for important intellectual content and approved the version.

Declaration of interests

A-WC (Chair), SA, CMT, and LH are members of the WHO Advisory Group. LMA is former Manager of the Australian Clinical Trials Registry and former member of the ICTRP Group. CMT is former Chair of the ISRCTN registry of LH is former Head of the Netherlands Trial Register. GK, JP, and VM declare no competing interests.

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